

COMPOSITE MATERIALS

IV B. Tech. I Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		C	CIE	SEE
A6ME41	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OVERVIEW:

This course is designed to provide a basic and important knowledge about the advanced class of materials, i.e. composites where its definitions, classification, fabrication, design and testing of composite materials are discussed. In the end, the current state-of-the-art research work happening in the field of multi-functional polymer nanocomposites, its preparation and applications are discussed.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Explain and differentiate composite materials based on specific applications.
2. Describe the various fabrication methods of polymer composites.
3. Choose suitable manufacturing route for generating metal and ceramic matrix composites.
4. Demonstrate various methods to test the Composites through NDT methods.
5. Elucidate the purpose and significance of Multi-Functional Composites.

UNIT-I	INTRODUCTION AND REINFORCEMENT MATERIALS
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Introduction: Definition, functions of matrix and reinforcement, composites merits and drawbacks, classification, polymer matrix composites, metal matrix composites, ceramic matrix composites, carbon-carbon composites, fibre reinforced composites and natural composites.

Reinforcement Materials: Glass, Carbon, Kevlar and Boron fibres.

UNIT-II	POLYMER COMPOSITES: MANUFACTURING METHODS
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Thermoset Composites Manufacturing: Hand layup, Spray up processes, Resin Transfer Moulding (RTM), Filament winding, Pultrusion, Prepreg Technique, Automated Fibre Placement (AFP) and Automated Tape Laying (ATL).

Thermoplastic Composites Manufacturing: Thermoplastic tape winding, Hot press technique, Diaphragm forming process, Injection moulding.

UNIT-III	METAL AND CERAMIC COMPOSITES: MANUFACTURING METHODS
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Metal Matrix Composites: Solid state methods - hot isostatic pressing (HIP), Foil diffusion bonding. Liquid state methods - Stir casting, Squeeze casting, Centrifugal casting, Melt infiltration.

Ceramic Matrix Composites: Sintering and CVD.

Manufacturing Process Selection Criteria: Production Rate/Speed, Cost, Performance, Size, Shape.

UNIT-IV	COMPOSITES DESIGN AND NDT
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Composites Design: Laminate theory, Rule of mixtures, symmetry and balance.

Non-Destructive Testing (NDT) of Composites: Visual inspection, Tap testing, Ultrasonic inspection, X-ray inspection, Thermography.

UNIT-V	MULTI-FUNCTIONAL POLYMER NANOCOMPOSITES
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Carbon Nanotubes (CNTs) based Multi-Functional Polymer Nanocomposites: Introduction and necessity of multi-functional composites, CNTs based multi-functional composites fabrication methods: Ultrasonication, In-Situ method, Grafting CNTs onto fibre surfaces - spraying method, and chemical grafting. Properties and Applications of CNTs based multi-functional composites.

Graphene based Multi-Functional Polymer Nanocomposites: Introduction, properties and applications.

Text Books:

1. Sanjay K. Mazumdar, Composites Manufacturing: Materials, Product, and Process Engineering, CRC press, 2001, ISBN: 0-8493-0585-3.
2. M. Balasubramanian, Composite materials and processing, CRC press, 2013, ISBN: 9781439880548.
3. Madhujit Mukhopadhyay, Mechanics of Composite Materials, Universities Press, 2005, ISBN: 9788173714771.
4. Ashutosh Tiwari and S. K. Shukla, Advanced Carbon Materials and Technology, Wiley, 2014, ISBN: 9781118686232.

Reference Books:

1. Flake Campbell Jr, Manufacturing processes for advanced composites, Elsevier, 2003, ISBN: 9781856174152.
2. Ehsan Bafekrpour, Advanced Composite Materials: Properties and Applications, De Gruyter Open Poland, 2017, ISBN: 9783110574401.